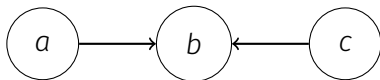




Bayesian networks vs factorisations

How can we factorise $p(a, b, c)$?



green: $p(a, b, c) = p(a)p(b)p(c)$

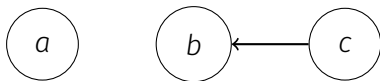
yellow: $p(a, b, c) = p(a)p(b|a)p(c|b)$

pink: $p(a, b, c) = p(b)p(a|b)p(c|b)$

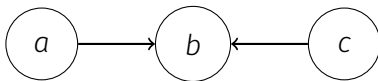
orange: $p(a, b, c) = p(a)p(b|a, c)p(c)$

Which Bayesian network(s) corresponds to $p(a, b, c) = p(a)p(c|b)p(b)$?

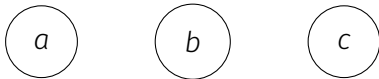
yellow



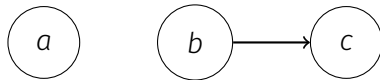
green



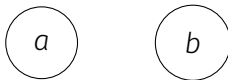
pink



orange



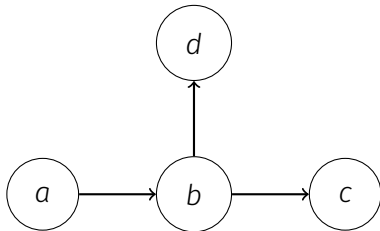
- A Bayesian network (BN) illustrates
 1. one possible factorisation
 2. certain conditional independencies (unless fully connected).
- **Example:** if $p(a, b)$ factorises according the BN below,



it holds that

1. $p(a, b) = p(a)p(b)$
 2. and, consequently, $p(b|a) = p(b)$ and $p(a|b) = p(a)$.
- A useful strategy when using BNs:
 - use the BN to factorize the joint distribution of all variables,
 - derive properties of interest from this factorisation.

How can we factorise $p(b, c, d)$?

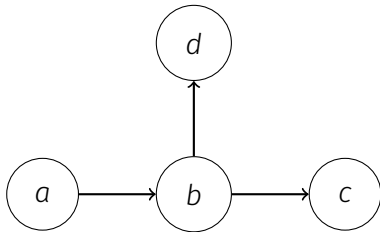


green: $p(b, c, d) = p(b)p(b|c)p(c|d)$

yellow: $p(b, c, d) = p(d)p(c|d)p(b|c)$

orange: $p(b, c, d) = p(b)p(c|b)p(d|b)$

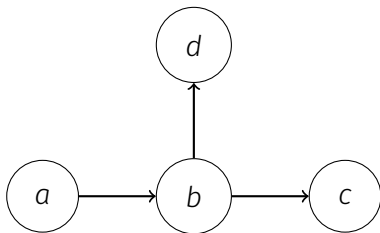
- We can follow the strategy mentioned earlier
 - use the BN to factorize the joint distribution of all variables,
 - derive properties of interest from this factorisation.



- According to the BN

$$p(a, b, c, d) = p(a)p(b|a)p(c|b)p(d|b)$$

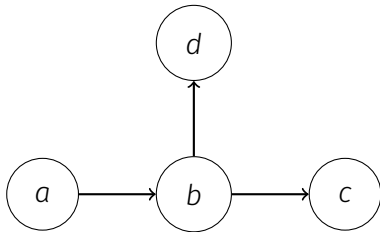
- We can follow the strategy mentioned earlier
 - use the BN to factorize the joint distribution of all variables,
 - derive properties of interest from this factorisation.



- According to the BN

$$\begin{aligned} p(a, b, c, d) &= p(a)p(b|a)p(c|b)p(d|b) \\ \Rightarrow p(b, c, d) &= \int p(a)p(b|a) da p(c|b)p(d|b) \\ &= p(b)p(c|b)p(d|b) \end{aligned}$$

- We can follow the strategy mentioned earlier
 - use the BN to factorize the joint distribution of all variables,
 - derive properties of interest from this factorisation.



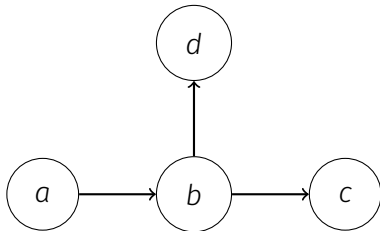
- According to the BN

$$p(a, b, c, d) = p(a)p(b|a)p(c|b)p(d|b)$$

$$\Rightarrow p(b, c, d) = \int p(a)p(b|a) da p(c|b)p(d|b) \\ = p(b)p(c|b)p(d|b)$$

Conclusion: the a variable and its node can be ignored here.

How can we factorise $p(a, b, c)$?



green: $p(a, b, c) = p(a)p(b|a)p(c|b)$

yellow: $p(a, b, c) = p(b)p(a|b)p(c|b)$

orange: $p(a, b, c) = p(a)p(c|a)p(b|c)$